IsDB-BISEW : Islamic Development Bank Bangladesh Islamic Solidarity Educational Wakf

IT SCHOLARSHIP PROGRAMME

Non-Profit Organization & Blood Donation Management System

Course Name: Web Application Development using ASP.NET
Module Name: SQL Server 2022
TSP: PeopleNTech Institute of Information Technology

Submitted To:	**Submitted By:**
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1. **Project Foundation and Vision**

1.1 Organizational Context

Foundation: The "Baro Baishdia Village Development Society" was established on March 2, 2022, in Barisal, Bangladesh. Founded by Shamsul Arefin, the organization operates with a workforce of 50 employees and a significant yearly budget of 10,000,000.00 BDT. The primary mission is to manage diverse community development projects and provide a critical lifeline through organized blood donation services.

Vision: Our vision is to build a self-reliant and resilient community where every individual has access to essential services and a dignified life. We strive to bridge the gap in rural development through diverse community projects and ensure that no life is lost due to a lack of blood, making organized blood donation a core pillar of our humanitarian mission.

1.2 The Technological Problem

Before the implementation of NPOBDMS, the organization faced "data silos"—information about donors, financial contributions, and blood requests were kept in disparate spreadsheets. This led to:

- **Response Latency:** Delays in matching patients with blood donors.
- **Financial Opacity:** Lack of automated auditing for donations.
- **Resource Mismanagement:** Inability to track volunteer engagement and project milestones.

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2. Database Architectural Design (DDL)

The database, named NPOBDMS, is designed for high availability and integrity. It utilizes specific file growth settings, with a data file (.mdf) and a log file (.ldf) optimized for transactional performance.

2.1 Core Infrastructure Tables

The system relies on a set of foundational tables that categorize stakeholders and logistical data:

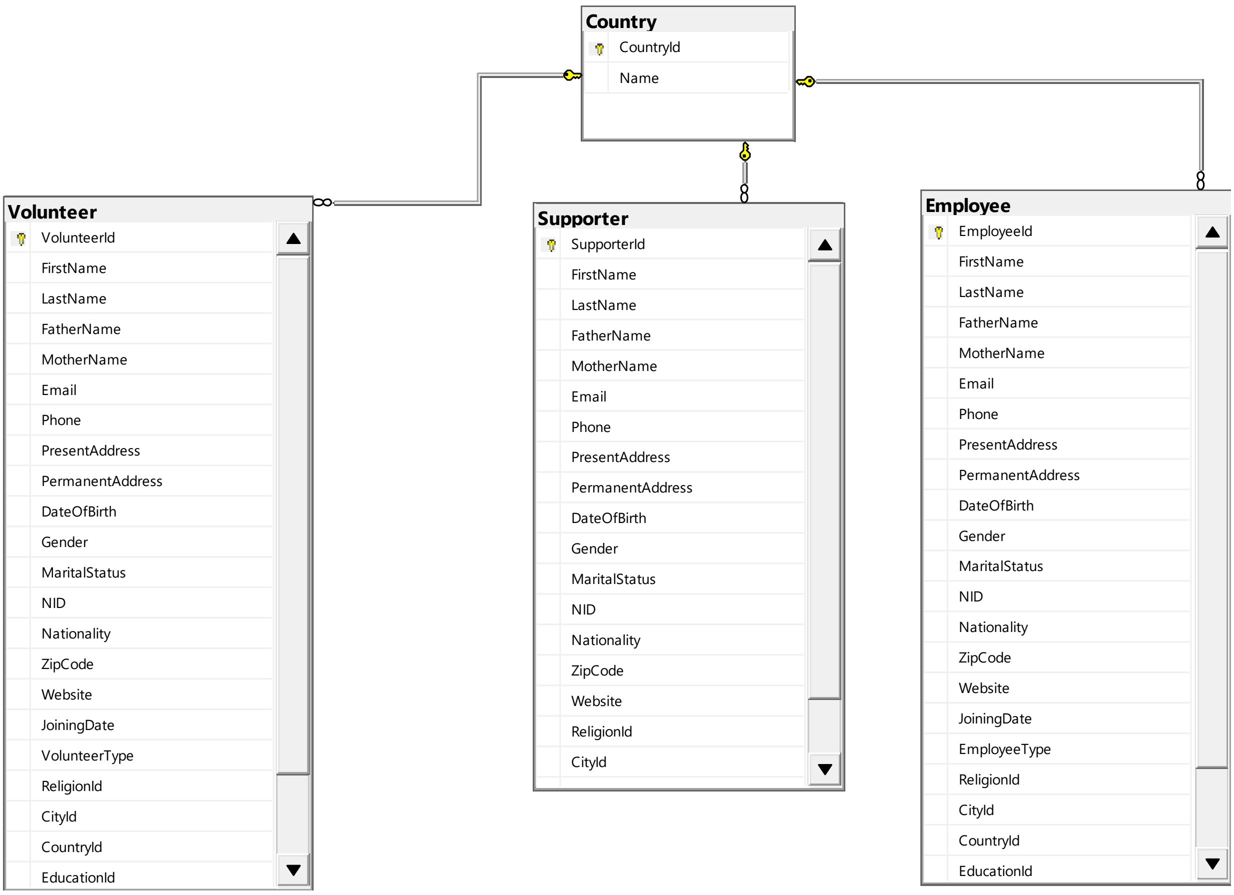
- **Organization:** Stores high-level administrative details, founder information, and financial health.
- **Geographic Hierarchy:** The Country and City tables ensure standardized address entry, preventing data duplication.
- **Demographics:** Tables for Religion and BloodGroup provide the necessary filters for targeted social and medical services.

2.2 Human Capital Management (HCM)

The schema distinguishes between different types of human resources:

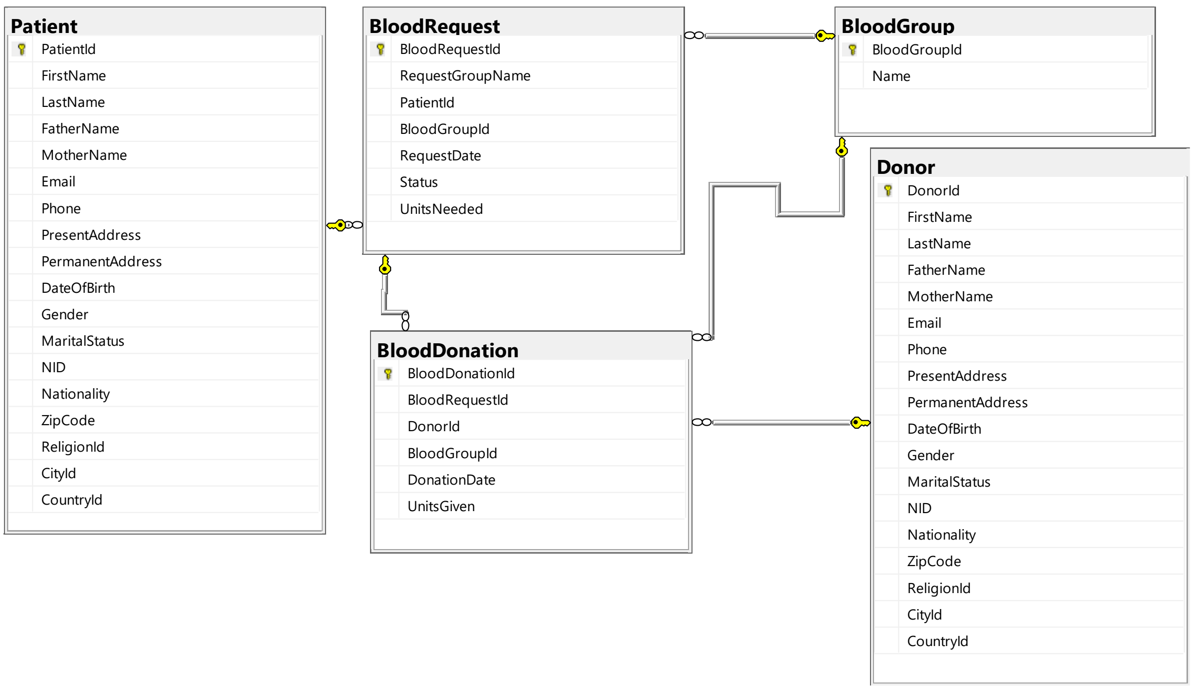
- **Employee Table:** Tracks permanent, contractual, and probation staff with attributes for education, designation, and institute.

- **Volunteer Table:** Includes joining dates and availability types (Full-Time, Part-Time, Occasional).
- **Supporter Table:** Tracks external advocates and their corporate affiliations via the Company table.



2.3 Medical & Financial Integration

- **Donor & Donation:** Every donor is linked to a unique ID, and every transaction is recorded with a payment method (Cash, Bank, Mobile Banking, etc.) and a specific amount.
- **Patient & Blood Request:** Patients are registered with full demographic details, and their blood requests are tracked by units needed and status (Pending, Fulfilled, and Cancelled).
- **Blood Donation:** This junction table bridges the gap between Blood Request and Donor, recording the exact units given and the date of donation.



3. Data Integrity and Business Rules

To ensure "Real-Life" reliability, the system implements several layers of constraints:

- **Primary Keys & Identity:** Every table uses an auto-incrementing IDENTITY(1,1) primary key to ensure unique record identification.
- **Check Constraints:**
    - **Phone Validation:** Ensures phone numbers follow the 11-digit Bangladeshi format (starting with 01).
    - **Email Validation:** Enforces the presence of standard email characters (@ and .).
    - **Financial Integrity:** All budget and donation amounts must be non-negative.
    - **Medical Integrity:** Limits Blood Group names to the eight standard biological types.
- **Default Values:** Nationality defaults to 'Bangladeshi' and dates default to the current system time (GETDATE()).

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4. Advanced Programmability

The NPOBDMS is more than a storage unit; it is an active application backend.

4.1 Stored Procedures (SP) for Automation

The system uses stored procedures to handle complex operations securely:

- **spInsertDonor:** Standardizes the onboarding of new donors by requiring essential demographic and contact info.
- **spGetDonorsByCity:** Allows rapid retrieval of local donors during emergencies.
- **spUpdateDonation:** Handles financial adjustments while maintaining data consistency.

4.2 Intelligence through Functions

- **fnGetTotalDonation:** A scalar function that calculates the total lifetime value (LTV) of a specific donor within the last year.
- **fnGetDonationSummary:** A multi-statement table-valued function that generates a monthly breakdown of total amounts and donation counts for any given year.

4.3 Trigger-Based Auditing

To maintain the trust of donors and regulators, the system features automated auditing:

- **Donation Audit:** The trAfterDonationUpdate trigger logs every change to a donation record, capturing the old amount, new amount, and timestamp into a separate DonationAudit table.
- **Real-time Validation:** The trCheckDonationAmount trigger prevents any record from being saved if the amount is zero or negative, providing a final layer of protection against manual data entry errors.

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5. Operational Scenarios (DML Case Studies)

Using the provided DML scripts, we can analyze how the system handles real-world NGO workflows.

5.1 Scenario A: Emergency Blood Procurement

When a patient at a medical camp requires blood, the coordinator executes:

1. **Search:** Queries vBloodDonationSummary to see current stock/history of blood groups.
2. **Match:** Uses fnGetDonorsByBloodGroup to find active donors who have given that specific group before.
3. **Contact:** Retrieves phone numbers from the Donor table using IX_Donor_Phone for instant access.

5.2 Scenario B: Financial Reporting for Stakeholders

At the end of the fiscal year, the board requires a "Donation Report":

- The system uses a **LEFT JOIN** between the Donor and Donation tables to identify all contributors, including those who may not have donated recently.
- **ROLLUP and CUBE** functions are applied to provide subtotals by year, month, and payment method, allowing management to see that "Mobile Banking" might be the most popular method for small-scale donors.

5.3 Scenario C: Performance Benchmarking

Management uses **Ranking Functions** to identify top performers:

- **DENSE_RANK()** is used to rank donors by the total amount contributed, allowing the NGO to send personalized "Thank You" messages to its top 10% of supporters.
- **NTILE(3)** divides donors into tiers (High, Medium, Low) for targeted fundraising campaigns.

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6. Technical Maintenance and Administration

6.1 Database Deployment

To deploy the system, administrators must ensure the directory C:\Program Files\Microsoft SQL Server\MSSQL17.MSSQLSERVER\MSSQL\DATA\ exists or modify the CREATE DATABASE paths to match the target server.

6.2 Performance Optimization

The system is pre-configured with **Non-Clustered Indexes:**

- IX_Supporter_Email and IX_Donor_Phone ensure that lookups for contact information do not require a full table scan, maintaining high performance even as the database grows to thousands of records.
- IX_Donation_Date allows the vDonationReport to render quickly during high-traffic auditing periods.

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7. Future Expansion & Scalability

While the current NPOBDMS is robust, the case study identifies three areas for future growth:

1. **Image Integration:** The existing Image Table is prepared to store file paths for ID photos and event documentation, which can be integrated into a web-based portal.
2. **AI-Predictive Modeling:** By analyzing the Blood Request history, future updates could include machine learning to predict seasonal blood shortages.
3. **Global Reach:** While currently focused on Bangladesh, the Country table allows for seamless expansion into international operations.

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8. Conclusion

The NPOBDMS project provides a scalable, secure, and highly automated framework for non-profit management. By centralizing stakeholder data, medical requests, and financial transactions into a single relational schema, the "Baro Baisha Village Development Society" can now operate with full transparency and maximum efficiency, ultimately saving more lives through better-coordinated blood donations.

Thank You!!